

Multivariate Encryptions with LL' perturbations

- Is it possible to repair HFE in encryption? When 0 makes a difference-

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Abstract. We will present here new multivariate encryption algorithms. This is interesting since few multivariate encryption schemes currently exist, while there exist many more multivariate signature schemes. Our algorithms will combine several ideas, in particular the idea of the LL' perturbation originally introduced, but only for signatures, by Gouget and Patarin in Vietcrypt 2006. In this paper, the LL' perturbation will be used for encryption and will differ greatly from the paper by Gouget and Patarin. As we will see, our algorithms resist all known attacks (in particular Gröbner attacks and MinRank attacks) and have reasonable computation times.

Keywords: Post-Quantum public key encryption · Post-Quantum key exchange · Multivariate cryptography · HFE

1 Introduction

Multivariate Cryptography is one of the main families of algorithms used to design post-quantum public key cryptography. We can trace its origin with the article [MI88] that introduced the so-called C^* scheme. The scheme was broken in [Pat95] but it led to multiple algorithms. With multivariate cryptography it is supposedly possible to perform encryption, signatures and authentication. Yet, in recent years almost all known multivariate encryption algorithms have been broken due to the discovery and improvement of attacks such as MinRank attacks. Due to this, at present most of the algorithms that are based on the MQ problem can only be used for signatures or authentication. For example the UOV [KPG99] family is currently largely represented in the recent NIST standardization process.

Another major multivariate algorithm is Hidden Field Equations (HFE) [Pat96]. Until recently HFE was one of the main candidate for multivariate cryptography, especially since it was one of the few able to be used for encryption (contrary to UOV mentioned previously) as well as signatures. Since its origin it was suggested to add perturbations in order to reinforce its security.

2. VANILLA HFE

HFE has a lot of variants. In the recent article [CMRPV24] the most important perturbations are presented with their current cryptanalysis properties. In [CMRPV24], the authors showed that it is hard to use HFE for encryption, indeed the perturbations that they studied have a high cost in order to avoid all attacks. However, the authors of [CMRPV24] have proposed a variant that seems to work for signatures. Other attempts around HFE or more generally around multivariate cryptography in encryption have been made cf. for example [SDP16], [IPS⁺18], [BM09], [CBD⁺09]. They generally have failed to resist all of the attacks [CCFST24], [BM09] or have been found to be far less effective than expected. This is why it is considered to be much easier to make multivariate signature schemes than encryption schemes.

The main novelty of this article is that we will be using the LL' perturbation introduced in [GP06] in a specific and original way for encryption. Our LL' can also be viewed as a particular case of the other perturbation $\hat{+}$ [FmRPP22], but as we will see, our LL' is much more efficient than $\hat{+}$.

In this paper, we will focus on studying the behaviour of LL' with HFE. However, this perturbation could be used with other multivariate scheme.

1.1 Related work

An attack against a previous version of our scheme has been published in [STV25]. The attack consisted in identifying the ciphertexts that were successfully decrypted in order to obtain information on the kernel of the linear form involved in our variant. Afterward the authors were able to get rid of them and perform an MinRank attack. However in our new protocol it is not possible to recover the messages that were successfully decrypted, as explained in Section 3.6, which mitigates the attack.

2 Vanilla HFE

Here by “vanilla” HFE or “Nude” HFE we mean the basic HFE scheme, *i.e.* without any additional perturbation.

The main idea for algorithms based on the difficulty to solve a quadratic system (Multivariate Quadratic problem) is to find structured quadratic polynomials that make the problem easy to solve. Then, we hide their structure via secret linear maps. For example let F be a quadratic polynomial such that the equation in x , $F(x) = y$ is easy to solve. Such polynomial F will be called the central map of the system. Let T, S be two bijective linear maps. The public key will be $T \circ F \circ S$, that way the public key will hopefully mimic the behaviour of a random quadratic map. Here the private key will be: T, S, F , hence with the private key and the hypothesis that F is easily “invertible” we can make an encryption/signature algorithm.

HFE uses such structure. Let q be a power of a prime number and d, n two integers, HFE utilizes the fact that the field $\mathbb{F}_{q^n}$ can be seen as the vector space

$\mathbb{F}_q^n$. Then an univariate polynomial $F(x)$ of $\mathbb{F}_{q^n}$ can also be seen as a multivariate polynomial $F(x_1, \dots, x_n)$. For vanilla HFE $F(x)$ will be a polynomial like this:

$$F(x) = \sum_{i,j \leq d} a_{ij} x^{q^i + q^j}$$

We see that the monomials are quadratic in $\mathbb{F}_q$ (x^{q^i} is linear in $\mathbb{F}_q$ as power of the Frobenius morphism, hence $x^{q^i + q^j} = x^{q^i} x^{q^j}$ is a quadratic term) and that the degree is limited by $D = 2 * q^d$. This is to ensure that Cantor-Zassenhaus [CZ81] algorithm can find a solution in a reasonable time. Then let T, S be two bijective linear maps $\mathbb{F}_q^n \rightarrow \mathbb{F}_q^n$. Let $\phi : \mathbb{F}_{q^n} \rightarrow \mathbb{F}_q^n$ be an isomorphism from the field extension of $\mathbb{F}_q$ to the vector space. Then the public key will be:

$$T \circ \phi \circ F \circ \phi^{-1} \circ S.$$

However most of the time in literature the isomorphism ϕ is not written but only implied. So the public key could also be written as:

$$T \circ F \circ S.$$

The private key will be:

$$T, S, F.$$

Due to the fact that the central map is quadratic there exist a matrix representation. This representation is used in the implementation of HFE and in the cryptanalysis. Hence, when there's no ambiguity we will note the corresponding matrix representation in bold with the same letter. For example, the matrix representation of F will be $\mathbf{F}$ (size $n \times n$) and the matrix representation of S and T will be $\mathbf{S}, \mathbf{T}$. In our case

$$\mathbf{F} = \begin{pmatrix} \mathbf{A} & 0 \\ 0 & 0 \end{pmatrix}$$

where $\mathbf{A}$ is a matrix of size $d = \log_q(D)$. Hence, the matrix representation of F is sparse.

From an attacker's point of view the goal would either be to invert the system directly (via Gröbner basis for example) or recover the linear map T or S (via a MinRank attack for example). Both will be using the fact that the degree of the central map F is limited.

When the level of security is higher than 80bits, vanilla HFE requires large D , therefore unrealistic computation time. This is why we need perturbations from the main algorithm in order to achieve higher level of security.

Here, we will only describe how HFE works in encryption mode but it could also be used in signature.

- Let Alice be the one who sends a message and Bob the one who receives it.
- Let m be the message, or a session key that Alice wants to send encrypted to Bob.
- Let $P = T \circ F \circ S$ be the public key of Bob.

2. VANILLA HFE

- Alice sends $m' = P(s||m)$ to Bob where s is a salt.
- Bob solves in x the equation $P(x) = m'$.
- For this Bob uses his private key to obtain the equation $f(S(x)) = T^{-1}(m')$.
- F being a univariate polynomial of $\mathbb{F}_q^n$ he uses for example Cantor-Zassenhauss or Berlekamp algorithm to solve the equation to find $S(x)$ and then x and then m since $x = s||m$.

Remark 1. As P is a system of multivariate polynomial in $X_1, \dots, X_n$ $P(X_1, \dots, X_n) = P_1(X_1, \dots, X_n) \dots P_n(X_1, \dots, X_n)$. Let $x \in \mathbb{F}_q^n$ The notation $P(x)$ means $P(x) = P_1(x_1, \dots, x_n) \dots P_n(x_1, \dots, x_n)$. We will use this notation in the rest of the article.

2.1 HFE $\hat{+}$

We will briefly present here the variant $\hat{+}$ of HFE. This variant is key to understand ours, as both are very similar in their effects. However we will later show that our variant becomes more effective because we change the method of decryption.

As with vanilla HFE we define

$$F(x) = \sum_{i,j \leq d} a_{ij} x^{q^i + q^j}$$

Let t be the $\hat{+}$ parameter and let $p_1, \dots, p_t$ be random quadratic forms i.e $p_i : \mathbb{F}_q^n \rightarrow \mathbb{F}_q : \sum_{j,k \leq n} a_{i,j,k} x_j x_k$. We can use the trace to give a univariate form hence we define $\tilde{p} : \mathbb{F}_{q^n} \rightarrow \mathbb{F}_q : \text{Tr}(\sum_{j,k \leq n} \alpha_{i,j,k} x^{q^k + q^j})$. Hence, we define the central map of HFE $\hat{+}$ as $H = F + \sum_i r_i \tilde{p}_i$ where $r_i \in \mathbb{F}_{q^n}$.

The public key is defined in a similar fashion as

$$P = T \circ H \circ S.$$

with S and T defined in Section 2

Decryption Protocol Recall the previous protocol

- Let Alice be the one who sends a message and Bob the one who receives it.
- Let m be the message, or a session key that Alice wants to send encrypted to Bob.
- Let $P = T \circ H \circ S$ be the public key of Bob.
- Alice sends $m' = P(s||m)$ to Bob where s is a salt.
- Bob uses his private key to obtain the equation $H(S(x)) = T^{-1}(m')$
- Here Bob cannot directly invert with Cantor-Zassenhauss as the degree of the $\tilde{p}_i$ is too high.
- $\tilde{p}_i$ can take at most q values hence Bob makes an exhaustive search on the possible values of the $\tilde{p}_i$ and fixes $\tilde{p}_i = \beta_i$.

- Bob solves $F(S(x)) = T^{-1}(m') - \sum_i r_i \beta_i$. If he made the right hypothesis this equation can be solved by Cantor-Zassenhauss as the degree is now bounded by d otherwise he continue the exhaustive search.
- Then Bob keeps the solution x where $\tilde{p}_i(x) = \beta_i$.

In other words, in order to decrypt one has to make an exhaustive search on the values of the $\tilde{p}_i$. Hence the cost of decryption of HFE $^+$ is q^t times the cost of decryption of HFE.

3 LL' perturbation

3.1 Our LL' perturbation in encryption

Let $P_1, \dots, P_n$ be the public key equations corresponding to a multivariate central map (e.g. HFE). $P_1, \dots, P_n$ are quadratic polynomials in $(X_1, \dots, X_n)$. Let r be an integer such that $1 \leq r \leq n$. Let L_i , $1 \leq i \leq r$ and L'_i , $1 \leq i \leq r$ be $2r$ secret linear forms in $x_1, \dots, x_n$. Our “outside” LL' consists in adding secret linear combination $\sum_i \alpha_{i,j} L_i L'_i$ to the equations P_j , $1 \leq j \leq n$.

Therefore $P'_j = P_j + \sum_i \alpha_{i,j} L_i L'_i$.

Then for our LL' variant the public key will $P'_1, \dots, P'_n$ while $P_1, \dots, P_n$ will remain secret.

3.2 Analog description

Let $P_1, \dots, P_n$ be the public key that correspond to a multivariate central map (in our case HFE). Let S and T be (as above) the two secret linear transformations. Let $\tilde{F}_1, \dots, \tilde{F}_n$ be the secret quadratic polynomials used for decryption. For example for HFE, $\tilde{F}_1, \dots, \tilde{F}_n = \phi \circ F \circ \phi^{-1}$. $\tilde{F}_1, \dots, \tilde{F}_n$ are the secret polynomials in $Z_1, \dots, Z_n$ and $S(X_1, \dots, X_n) = (z_1, \dots, z_n)$. Our “inside” LL' consists in adding secret linear combinations to the A_j , $1 \leq j \leq n$ so $\tilde{H}_j = \tilde{F}_j + \sum_i \alpha_{i,j} L_i L'_i$.

In fact our “inside” and “outside” transformation are similar (since we can move from one to the other description by just modifying S and T), but for the analysis we will sometimes alternatively use “inside” or “outside”.

3.3 Decryption Protocol

The main idea of the LL' perturbation is that over $\mathbb{F}_2$ we have a probability $1/4$ that $L_i L'_i \neq 0$. In order to decrypt, we will evaluate the expected number of products $L_i L'_i \neq 0$ (it will be $r/4$ on average, but it can be much less as we will see below). Then, after exhaustive search on the indices i such that $L_i L'_i \neq 0$ i.e. $L_i L'_i = 1$ we will solve the system like a normal HFE system.

Recall the previous notations, we consider the Decryption protocol for our HFE LL'

- Let Alice be the one who sends a message and Bob the one who receives it

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- Let m be the message, or a session key that Alice wants to send encrypted to Bob.
- Let $P = T \circ H \circ S$ be the public key of Bob
- Alice sends multiple $m'_j = P(s||m_j)$ ($1 \leq j \leq k$) to Bob where s is a salt (we will later discuss the values required for the number k).
- Bob select an indice j and tries to solve in x the equation $P(x) = m'_j$.
- For this Bob uses his private key to obtain the equation $H(S(x)) = T^{-1}(m'_j)$.
- Here Bob cannot directly invert with Cantor-Zassenhauss as the degree of the $\tilde{p}_i$ is too high.
- Bob makes an exhaustive search fixes b_k indices $i_1, \dots, i_{b_k}$ and fixes the value of the corresponding $L_{i_j} L'_{i_j}$ as $\beta_{i_j} \in \mathbb{F}_q$. All other $L_i L'_i$ are supposed to be 0.
- For each guesses, Bob tries to solve $F(S(x)) = T^{-1}(m'_j) - \sum_{i_j}^{i_k} \alpha_{i_j} \beta_{i_j}$ if the hypothesis is right then this equation can be solved efficiently using algorithms such as Cantor-Zassenhauss as the degree is now bounded by d .
- Then Bob keeps the solution x such that $P(x) = m'_j$ (in other words those that correspond to the hypothesis made by bob).
- if it doesn't work then Bob asks Alice for a new set of message.

In our scheme we bound the exhaustive search hence we are able to be much more efficient than $\hat{+}$. Overall $\hat{+}$ makes an exhaustive search on all of the q^t possible values while we require an exhaustive search on q^{b_k} values with $b_k < r$. We require to send far more messages however. The discussion on the number of messages required to be sent and the appropriate bound will be answered in section 5.

3.4 Variants

- In this paper, $x_1, \dots, x_n$ will be over the small field GF(2). Yet, this perturbation LL' can work on other very small Field (GF(3) for example) however it will be less efficient as the probability to cancel a linear form will decrease.
- Usually, we will have public equations of degree 2. In appendix we will also present a variant with equations of degree 3.
- It is also possible to use very simple linear combination for $L_i L'_i$. Instead of a general linear combination of the form: $\sum_i \alpha_{i,j} L_i L'_i$ we can imagine to just add $L_i L'_i$ to P_i . At present, this variant does not seems to reduce the security significantly, but it does not accelerate the decryption either (since the most difficult part of the decryption in the inversion of f and the fact that we have to repeat this operation multiple times).
- In this paper, we will use our LL' perturbation with HFE central map. Other central maps may also be used, for example C^* , cf. [MI88]. However, so far all attempts to repair C^* have failed. Therefore we are more confident with HFE and a specific analysis will probably be required on C^* .

Remark 2. A central map (as HFE for example) is really needed, *i.e.* LL' cannot be used alone. This is because a random linear combination of LL' equation will have a non-negligible probability to have a small rank and from this property it is possible to find **S** or **T**.

These variants are only general development ideas and have not been subject to a full extent analysis. Hence they will not be part of the final specification.

3.5 How to make our LL' less expensive in computations

As said above, about $r/4$ products $L_i L'_i$ are non zero. However sometimes much less products are non zero. In order to be in such a good situation for decryption, one possibility is to send many messages or session keys and hope that at least one of them has only a small number of $L_i L'_i \neq 0$. We will see below (sections 5 and 6) how many messages to send on average in order to have a good probability for this to occur.

3.6 Transmission of a session key

Usually a public key encryption algorithm is used not to encrypt a plaintext message, but to encrypt a session key that will be used later. Then we can imagine that (with the same public key) we will encrypt potential session keys $k_1, \dots, k_l$ such that from any of these value k_i it is possible to find all the other values k_j (simple transformation can be used to do so). The receiver will decrypt one of the keys k_α . This is not yet the chosen session key; it is only meant to serve as an intermediate value. Otherwise, it could leak informations on the number of $L_i L'_i = 0$ obtained with this key. A great number of informations like this could lead to the discovery of the kernel of the $L_i L'_i$, which in turn could allow an opponent to use a projection to make the perturbation vanish and perform a regular attack. More details on how to exploit this leakage can be found in [STV25].

To prevent such information leakage, the recipient uses the key k_α to recompute all the keys $k_1, \dots, k_l$. He then choses at random one of the key k_i and he will send back a hash of k_i (or alternatively, he will use a key derived by k_i and send an encrypted text with this key). He will also check that the message that were sent were well computed, to do so he select some of the messages randomly and check their validity. This mechanism ensures that the sender does not know which ciphertext was decrypted, and prevents any correlation between ciphertexts and successful decryption.

The protocol is constructed in the following way:

- Alice wants to send the a session key K to bob.
- Alice generates K_o at random, then she generates k keys of the form $K_i = \sigma^i(K_o)$ where σ is a public permutation.
- Alice sends Bob, messages of the form $M_i = (P(K_i), i)$.
- Bob decrypts one of the message K_j .
- Bob recomputes the messages and selects at random some of the messages sent and checks that Alice respected the protocol.
- If the protocol is respected then Bob select one of the messages K_i and sends i to Alice.
- The session key is now K_i .

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When Bob checks if Alice respected the protocol he only needs to encrypt back the messages and check if their encryption is compatible with the messages sent by Alice. When the number of messages is low (less than 100) he can chose to check them all. However for higher numbers of messages he can chose to encrypt k of them.

Let's evaluate this number. Let p be the proportion of faulty messages, k the number of messages that Bob will randomly check to have been well computed and m the number of messages that Alice sent. Then the probability for an attacker to obtain informations is $prob = p \prod_{i=0}^{k-1} \frac{m-mp-i}{m-i}$. In order to obtain significant information an attacker needs to obtain $d+1$ vectors where all of the LL' vanishes. But here the complexity will depend on the number of LL' that were allowed to be non zero in the decryption. If we allow no LL' to be non zero then we can try to guess a vector space where all LL' vanishes we only need to fraud $d+1$ times in a row and then perform a min rank attack c.f hybrid attack. The complexity would be $\frac{1}{prob^{d+1}}$.

But in our case for security above 128 bit we allow LL' to be non zero at most 1 time. Then in order to perform this attack and try to recover a vector space that will make some of the LL' vanish, he needs to make hypothesis on the right non zero LL' . Then the complexity would be $\frac{\tau}{prob^{d+1}}$.

Remark 3. If no sent message has a small number of $L_i L'_i \neq 0$ then the receiver will have to ask for a new set of messages. In order to not leak any information when this occur, it is possible to randomly ask for new messages even if we were able to decrypt the system.

3.7 Transmission of a specific message m

Although, public key encryption algorithms are usually used to exchange session keys, our algorithm can also be used to transmit a specific message m . This variant is generally less convenient but still feasible. It is mandatory that an opponent does not have much more equations on the cleartext than the number of variables of the cleartext (otherwise the system becomes much easier to solve with a Gröbner basis attack). However, to ensure that at least one ciphertext yields a small number of $L_i L'_i \neq 0$, multiple ciphertexts must be generated for the same message.

To prevent information leakage between these ciphertexts while using the same public key, we transform the message M using several encodings $M_1, \dots, M_k$ with the help of a public function (such as an AES); the role of this public transformation is to break linearity and prevent correlation between ciphertexts. For example, Let $K_1, \dots, K_k$ be keys of an AES-128, then $M_i = AES(K_i, M)$. Then, the M_i will be encrypted with the public key of our HFE LL' and the keys K_i will be made public. The properties of the AES makes the M_i indistinguishable from a random values. Therefore, no additional information is leaked by encrypting multiple variants of the same message. The protocol is constructed in the following way:

- Alice wants to send the message M to bob.
- She sends Bob k messages of the form $M_i = P(AES(K_i, M))$ where K_i are public and P is the public key of HFE LL'.
- Bob successfully decrypts one of the messages of indices j . Hence he recovers M .
- Now that he obtains the message M Bob can check that Alice respected the protocol and that the encryption of the other values were done correctly by Alice.

With this protocol, Alice cannot know which message Bob managed to decrypt. Thus cannot use that knowledge to attempt a key-recovery attack.

3.8 Differences between our LL' and the LL' of [GP06]

In [GP06] a perturbation LL' was also described. However, the description differed greatly from ours and it was used in signature rather than encryption. In [GP06] n products $L_i L'_i$ are used. They were simply added to the public key without any linear combination: $P'_i = P_i + L_i L'_i$. In this paper the main difference is that we only have r products $L_i L'_i$ and that

$$P'_j = P_j + \sum_i \alpha_{i,j} L_i L'_i$$

or

$$A'_j = A_j + \sum_i \alpha_{i,j} L_i L'_i.$$

The second important difference is that in [GP06] not all of the public equations have to be satisfied, unlike in this paper where all the equations are valid equations.

3.9 LL' as a particular case of $\hat{+}$

The perturbation $\hat{+}$ was introduced in [FmRPP22]. With $\hat{+}$ we have r random quadratic equations that are linearly combined on the public (or secret) quadratic equation. In LL' we combine r product $L_i L'_i$. Therefore LL' can be seen as a special case of $\hat{+}$.

However, our perturbation will be much less costly in terms of computation time than $\hat{+}$. This is because $L_i L'_i$ is 0 with a probability $3/4$ unlike $1/2$ for $\hat{+}$.

Recall all the previous notations, Instead of adding a linear combination of random quadratic form $\hat{p}_i(x)$ we will add linear combination of $L_i L'_i$ where : $L_i(x) = \sum_i a_i x^{q^i}$.

It shows clearly the link between the two perturbations. Again we see that $\hat{+}$ is more general than LL'. However we will later show that it does not appear to induce any weaknesses to LL' compared to $\hat{+}$. Indeed our tests confirmed that the cryptanalysis of both perturbations is very similar.

However, in section 4.5 we present a specific attack for LL' that is more effective than used on $\hat{+}$. Nevertheless, overall this new attack is not very efficient.

4 Cryptanalysis of LL'

4.1 General considerations

In this Section we will study how LL' resists against the classical known attacks in multivariate cryptography (such as MinRank and Gröbner attacks). We may be afraid by the fact that in our schemes, sometimes almost all, or even all, the terms $L_i L'_i$ are 0 on a given ciphertext. We will show that an attacker cannot easily exploit this property.

In MinRank attacks we study the public equations (including the hidden terms $L_i L'_i$) independently from its values, so the fact that some $L_i L'_i = 0$ does not intervene. In other words, MinRank targets the structure of the central map and especially rank defects but it does not target the values of the polynomials.

On Gröbner attacks we try to solve the public equations with all the terms $L_i L'_i$, and the fact that we want to decrypt even when all these terms are 0 does not really help since these terms are hidden. This was confirmed on many simulations that we did (cf. section 4.6).

Remark 4. It looks really difficult for an attacker to be able to recover the equations without the $L_i L'_i$ from cleartext/ciphertext pairs. This is because he does not know which message will be decrypted or which session key will be used (*i.e.* which messages or session keys with almost all $L_i L'_i = 0$).

We will now give more details about how LL' can resist all known attacks. Since LL' can be seen as a variant of $\hat{+}$ we will conduct a similar analysis as done in [FmRPP22] and compare $\hat{+}$ and LL'. First of all we will distinguish two types of attacks, Direct attacks (Gröbner Basis) and key recovery attacks (MinRank attacks).

The study of $\hat{+}$ as well as our simulations showed that LL' and $\hat{+}$ have a very similar behaviour against both types of attacks. So we will consider that both perturbations have the same impact on the degree of regularity of the polynomial. This is largely due to the fact that the evaluation of the degree of regularity is linked to the rank of the central map, and our tests showed that the ranks are equal. So it should follow the following formula:

$$d_{reg} = (d + r)/2 + 2$$

where r is the number of product of linear forms added to the public equations. As LL' is a variant of $\hat{+}$ we know that the previous formula is an upper bound of the degree of regularity.

For the MinRank attacks analysis we should distinguish two forms of MinRank that we will note MinRank **T** [BFP11], MinRank **S** [TPD21]. The goal of the MinRank will be to recover one of the linear map of the private key, once one of them is found the rest of the key is “easy” to find. Hence, MinRank **T** will denote the fact that we will try to recover the matrix **T** first and MinRank **S** the fact that we will try to recover the matrix **S**. Let us define first the MinRank problem.

Definition 1. Let $n, m, r, k \in \mathbb{N}$ and let $\mathbf{G}_1, \mathbf{G}_2, \dots, \mathbf{G}_k$ $n \times m$ be matrices over the field $\mathbb{F}$. The MinRank problem consists to find $u_1, u_2, \dots, u_k$ over $\mathbb{F}$ such that $\text{rank}(\sum_{i=1}^k u_i \mathbf{G}_i) \leq r$.

We need to rewrite the problem of finding one of the two matrices as a MinRank instance. The goal will be to use the public key matrices: $\mathbf{P}'_1 \dots \mathbf{P}'_n$ as the matrices $\mathbf{G}_1, \mathbf{G}_2, \dots, \mathbf{G}_n$ (N.B for MinRank $\mathbf{S}$ we will not directly use the public key matrices) and to use some of the entries of the matrices $\mathbf{T}$ or $\mathbf{S}$ as the solution vector.

We should also remind here the matrix writing of the HFE LL' map. Recall here that $H = F + Q$ where F is the central map of a HFE and Q the LL' modifier.

Lemma 1. Let $\mathbf{S}, \mathbf{T} \in M_{n \times n}(\mathbb{F}_q)$ be the linear map of the private key. Then the public key $\mathbf{P}'$ can be written

$$\mathbf{P}' = (\mathbf{P}'_1, \dots, \mathbf{P}'_n) = (\mathbf{S}\mathbf{M}_n\mathbf{H}^{*0}\mathbf{M}_n^t\mathbf{S}^t, \dots, \mathbf{S}\mathbf{M}_n\mathbf{H}^{*n}\mathbf{M}_n^t\mathbf{S}^t)\mathbf{M}_n^{-1}\mathbf{T}$$

where $\mathbf{H}^{*i}$ is the matrix representation of the q^i th power of the secret polynomial H and $\mathbf{M}_n$ denotes the matrix representation of the map ϕ .

4.2 MinRank $\mathbf{T}$

Let's start with MinRank $\mathbf{T}$, this method can be found in [BFP11].

Let q, n, D be standard HFE parameters, $(\mathbf{P}'_1, \dots, \mathbf{P}'_n)$ the public key and $\mathbf{T}, \mathbf{S}, \mathbf{H}$ the secret key as defined earlier. Then we have

$$(\mathbf{P}'_1, \dots, \mathbf{P}'_n) = (\mathbf{S}\mathbf{M}_n\mathbf{H}^{*0}\mathbf{M}_n^t\mathbf{S}^t, \dots, \mathbf{S}\mathbf{M}_n\mathbf{H}^{*n}\mathbf{M}_n^t\mathbf{S}^t)\mathbf{M}_n^{-1}\mathbf{T}.$$

So we can write

$$(\mathbf{P}'_1, \dots, \mathbf{P}'_n)\mathbf{T}^{-1}\mathbf{M}_n = (\mathbf{S}\mathbf{M}_n\mathbf{H}^{*0}\mathbf{M}_n^t\mathbf{S}^t, \dots, \mathbf{S}\mathbf{M}_n\mathbf{H}^{*n-1}\mathbf{M}_n^t\mathbf{S}^t).$$

We will write $\mathbf{U} = \mathbf{T}^{-1}\mathbf{M}_n$ and $\mathbf{W} = \mathbf{S}\mathbf{M}_n$. Then we have

$$(\mathbf{P}'_1, \dots, \mathbf{P}'_n)\mathbf{U} = (\mathbf{W}\mathbf{H}^{*0}\mathbf{W}^t, \dots, \mathbf{W}\mathbf{H}^{*n}\mathbf{W}^t).$$

Let $(u_{0,0}, u_{1,0}, \dots, u_{n-1,0})$ be the first column of $\mathbf{U}$ then we have

$$\sum_{i=0}^{n-1} u_{0,i} \mathbf{P}'_i = \mathbf{W}\mathbf{H}\mathbf{W}^t.$$

Due to the addition of random polynomials the rank of $\mathbf{H}$ is likely very high. However if we write these polynomials in the field extension then like for $\hat{+}$. There exist $\beta_1, \dots, \beta_r$ coefficient in $\mathbb{F}_{q^n}$ and $L_1, L'_1, \dots, L_n, L'_n$ linear forms, such that the central map $h(x)$ verifies the equation:

$$h(x) = f(x) + \sum_{i=0}^r \beta_i L_i L'_i$$

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where $f(x)$ is vanilla HFE central map. If we want to fully keep a univariate representation we can write, $L_i(x) = Tr(\sum_i a_i x^{q^i})$ (where Tr is the trace function of a field extension). Hence, there exist a linear map $\Pi : \mathbb{F}_{q^n} \rightarrow \mathbb{F}_{q^n}$ such that $\forall i, \Pi(\beta_i) = 0$. This map has a degree at least r [FmRPP22]. So we can further write:

$$(\mathbf{P}'_1, \dots, \mathbf{P}'_n) \mathbf{U} \Pi = (\mathbf{W} \mathbf{H}'^{*0} \mathbf{W}^t, \dots, \mathbf{W} \mathbf{H}'^{*n} \mathbf{W}^t).$$

where $\mathbf{H}' = \mathbf{H} \Pi$ where then the rank of the right member is at most $d + r$. Let $\mathbf{U}'$ be such that $\mathbf{U}' = \mathbf{U} \Pi$. Hence

$$\text{rank}(\sum_{i=0}^{n-1} u'_{0,i} \mathbf{P}'_i) = \log_q(D) + r$$

which is small so finding the first column of $\mathbf{U}$ reduces to solving a MinRank instance with $k = n$ and the rank is $\log_q(D) + r$ on the matrices $\mathbf{P}'_1, \dots, \mathbf{P}'_n$. As we can see the addition of r to the rank is due to the degree of the linear map Π . But the degree of this map is due to the randomness of the β_i (in the univariate representation of the perturbation), the fact that we took special quadratic polynomial in the form $L_i L'_i$ instead of random quadratic p_i (like in HFE $\hat{+}$) does not seems to change the complexity of the attack.

4.3 MinRank $\mathbf{S}$

MinRank $\mathbf{S}$ was first proposed by Tao *et al.* [TPD21].

Retaining the notations $\mathbf{U}$ and $\mathbf{W}$ from the previous attack, we have $(\mathbf{P}'_1, \dots, \mathbf{P}'_{n-1}) = (\mathbf{W} \mathbf{H}^{*0} \mathbf{W}^t, \dots, \mathbf{W} \mathbf{H}^{*n-1} \mathbf{W}^t) \mathbf{U}^{-1}$. Then we obtain

$$(\mathbf{W}^{-1} \mathbf{P}'_1 \mathbf{W}^{-1,t}, \dots, \mathbf{W}^{-1} \mathbf{P}'_{n-1} \mathbf{W}^{-1,t}) = (\mathbf{H}^{*0}, \dots, \mathbf{H}^{*n-1}) \mathbf{U}^{-1}.$$

This leads us to study the matrix whose line i is the first line of the matrix $\mathbf{H}^{*i}$ in other word it's the first line of the representative matrix of the central map after applying the Froebenius morphism i times. Let's call this matrix $\mathbf{G}$

We can decompose $\mathbf{H}$ in the following way: $\mathbf{H} = \mathbf{F} + \mathbf{Q}$. With $\mathbf{F}$ being the central map of a vanilla HFE and $\mathbf{Q}$ the perturbation $LL' / \hat{+}$. The Froebenius operation being linear we can study the effect of the operation on $\mathbf{F}$ and $\mathbf{Q}$ separately. The form of $\mathbf{F}$ is sparse then the matrix we will obtain will be of the

form:
$$\begin{pmatrix} \mathbf{A}_1 \\ 0 \\ \mathbf{A}_2 \end{pmatrix}$$

On the other the matrix by $\mathbf{Q}$ will be of no particular form: $\begin{pmatrix} \mathbf{q}_0 \\ \vdots \\ \mathbf{q}_n \end{pmatrix}$. However,

we can write that $\mathbf{q}_i$ is the first row of the matrix $\sum_{j=0}^t \beta_j^{q^i} \mathbf{Q}_j$. Indeed $\mathbf{Q}_j$ is the representative matrix of the polynomial $\hat{p}_j(x)$ or $Tr(L_j(x) L'_j(x))$ (respectively in the case of $\hat{+}$ and LL') whose image is in $\mathbb{F}_q$. Hence the polynomial is unchanged by the Froebenius morphism.

$$\mathbf{Z} = (\mathbf{U}^{-1})^t \times \mathbf{G}$$

$$\mathbf{Z} = (\mathbf{U}^{-1})^t \times \left(\begin{pmatrix} \mathbf{A}_1 \\ 0 \\ \mathbf{A}_2 \end{pmatrix} + \begin{pmatrix} \mathbf{q}_0 \\ \vdots \\ \mathbf{q}_n \end{pmatrix} \right)$$

It means that the rank of the right matrix is at most r and thus the rank of the matrix $\begin{pmatrix} \mathbf{A}_1 \\ 0 \\ \mathbf{A}_2 \end{pmatrix} + \begin{pmatrix} \mathbf{q}_0 \\ \vdots \\ \mathbf{q}_n \end{pmatrix}$ is at most $d + r$.

Furthermore, using this and the matrix equation of HFE.

Theorem 1. *Let $\mathbf{P}'_1, \dots, \mathbf{P}'_n$ be matrices of the public key and $\mathbf{W}$ the matrix previously defined. If one notes $(w_0^{-1}, w_1^{-1}, \dots, w_{n-1}^{-1})$ the first row of the matrix $\mathbf{W}^{-1}$, and $b_i = (w_0^{-1}, w_1^{-1}, \dots, w_{n-1}^{-1})\mathbf{P}_i$, then the matrix $\mathbf{V}$ whose rows are the b_i has a rank at most $d + r$.*

Proof. From the previous result we know that the rank of $\mathbf{V}\mathbf{W}^{-1^t}$ is bounded by $d + r$, hence the rank of $\mathbf{V}$ is bounded by $d + r$.

For both MinRank once we recover the entries of $\mathbf{T}$ or $\mathbf{S}$ we can recover of the totality of the key like in [BFP11], [TPD21].

As mentioned we have essentially reused the cryptanalysis of HFE $\hat{+}$ [FmRPP22]. We have successfully simulated this result using a modified version of the code that was published alongside the article [BBC⁺22]. We did not effectively perform the attack, in other word we did not effectively solve the MinRank. But we computed the rank that appeared in the equation previously mentioned, which confirms the analysis we have performed. So our results showed the augmentation of rank as we have announced.

We performed our tests on a toy example taking: $n = 20, d = 7, r = 1, 2, 3$ and using $\hat{+}$ with similar parameters $n = 20, d = 7, t = 1, 2, 3$.

In the end, we computed the rank of the matrix $\mathbf{V}\mathbf{W}^{-1^t}$ and the matrix $\mathbf{U}' = \mathbf{U}\mathbf{H}$. and found that their rank coincided with the theory we have presented.

4.4 Resolution of MinRank

The MinRank problem is solved using the support minor method as described in [BBB⁺22], [BBC⁺22]. The complexity of the resolution does not depend on the type of MinRank ($\mathbf{T}$ or $\mathbf{S}$) but only on the target rank, the Field characteristic and the degree of the extension.

$$\mathcal{O} \left((d+r)(n-1)^4 \binom{2(d+r)+1}{d+r}^2 \right).$$

This gives us a good idea of the number of LL' that needs to be added. We then have a least:

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$r = 12$ for 80 bits of security
 $r = 17$ for 100 bits of security
 $r = 24$ for 128 bits of security
 $r = 40$ for 192 bits of security
 $r = 56$ for 256 bits of security

(These numbers are obtained with the pessimistic assumption that $d = 0$ which is never the case, therefore in the parameters section these numbers will be slightly reduced).

4.5 Hybrid method

We can note that the fact that $L_i L'_i$ is more often 0 than a regular quadratic form Q_i it means that there are more linear map Π that cancels the perturbation. Hence, one could try to randomly compose the public key with a linear map Π_r (r stands for random) in hope that the perturbation LL' would vanish (or at least part of it).

Our attack works the following way: we select random linear map Π . Then we compose the public key P with Π in order to obtain $\tilde{P} = P \circ \Pi$. Then we try to obtain the private key of the system by solving the MinRank problem induced by $\tilde{P}$. We continue this process until we have obtained the private key.

If the projection Π cancels some of the $L_i L'_i$ it will induce a decrease of the rank. For example if one $L_i L'_i$ is cancelled it means that the rank of the first coordinate of $P(\Pi)T^{-1}$ is $d + r - 1$. And overall if we make k $L_i L'_i$ vanish the rank would become $d + r - k$.

Let's evaluate the number of tries required in order to remove at least one of the $L_i L'_i$. In other words, the probability to find Π such that their exist $1 \leq j \leq t$ such that $L_j(\Pi) = 0$ or $L'_j(\Pi) = 0$. It means that all the vectors of the basis of the image of Π are included in the kernel of L_j or L'_j . Let r be the rank of Π then the probability for Π to cancel $L_i L'_i$ is $p = 2/q^r - 1/q^{2r}$.

So the probability for Π to cancel at least k $L_i L'_i$ is $prob_k = \sum_{j=k}^t \binom{t}{j} p^j (1-p)^{t-j}$. In this case the target rank for the MinRank attack would become $d + r - k$. The rank of Π should be at least $d + r - k + 1$ so that we can obtain a rank defect.

So overall the complexity of the attack will become:

$$\mathcal{O} \left(\frac{1}{prob_k} (d + r - k)(n - 1)^4 \binom{2(d + r - k) + 1}{d + r - k}^2 \right).$$

We will show in Section 6 that this attack is less effective than the regular MinRank attacks indeed $prob_k$ decrease faster than the complexity of MinRank with a lower rank. This hybrid attack is however the only specific attack, that we know, where LL' is weaker than $\hat{+}$. As the number of cancelling of the polynomial (product of linear forms for LL' , random quadratic form for $\hat{+}$) we add is the only major structural difference with $\hat{+}$.

4.6 Direct attacks (Gröbner)

In this section we describe the behaviour of direct attacks using Gröbner basis against HFE LL'. To study the complexity of the attack, the good notion is the degree of regularity. So we will focus on the behaviour of the degree of regularity of HFE LL' central map. We will use the definition of Ding and Schmidt [DS13].

Definition 2. We define $B = \mathbb{F}[X_1, \dots, X_n] / \langle X_1^q, \dots, X_n^q \rangle$ and B_d its degree d subspace. Let P be a set of homogeneous polynomial $P = \{P_1, \dots, P_m\} \subset B_2^m$.

Let ψ_d be the map $\psi_d : B_d^m \rightarrow B_{d+2}$ defined as

$$\psi(b_1, \dots, b_m) = \sum_{i=1}^m b_i P_i$$

Then

$$R_d(P_1, \dots, P_m) := \ker(\psi_d).$$

Further let $T_d(P_1, \dots, P_m)$ be the subspace of trivial relations generated by the elements

$$\{b(P_i e_j - P_j e_i) | 1 \leq i < j \leq m, b \in B_{d-2}\},$$

and

$$\{b(P_i^{q-1})e_i | 1 \leq i \leq m, b \in B_{d-2(q-1)}\}.$$

Here e_i means the i -th unit vector consisting of all zeros except 1 at the i -th position $e_i = (0, \dots, 0, 1, 0, \dots, 0)$. The degree of regularity of a homogeneous quadratic set is then

$$Dreg(P_1, \dots, P_m) := \min\{d | R_{d-2}(P_1, \dots, P_m) / T_{d-2}(P_1, \dots, P_m) \neq \{0\}\}.$$

The work of Joux and Faugère [FJ03] showed the link between the rank and the degree of regularity. Analysis of $\hat{+}$ showed that the degree of regularity was increased by the action of $\hat{+}$ (cf. [FmRPP22]). Our tests confirmed this result and showed that LL' had a very similar impact. The computation of the Table

Number of perturbation	HFE $\hat{+}$	HFE LL'
0	0.86	0.86
1	1.75	1.65
2	2.17	2.13
3	2.11	2.14
4	2.09	2.09

Table 1. Time to solve the public key of HFE $\hat{+}$ compared with HFE LL' using a Gröbner basis

1 have been made on a toy example using $q = 3, D = 7, n = 10$. It shows that both of perturbations have a similar impact against Gröbner.

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Note that the results are the average for 10 Gröbner inversion so the fact that the time may slightly decrease is simply due to the usage of small parameters. We have a bound for the degree of regularity of HFE. Ding *et. al* [Kle12] showed that the degree of regularity D_{reg} is such that. $D_{reg} \leq \frac{(q-1)d}{2} + 2$. It leads to an evaluation of the complexity of the resolution of the Gröbner basis of the system.

In order to find a Gröbner basis for a system we are using algorithms from Jean-Charles Faugère (F4, F5 [Fau99][Fau02]).

As we saw LL' have a good impact against Gröbner attacks. However, we could wonder if there exists vulnerabilities that depends in the choice of the ciphertext. For example, let f be the public key of a HFE LL' and let $x = \{x_1, \dots, x_n\}$ and $Y = f(x)$ be such that $\forall i \leq r \ L_i(X)L'_i(X) = 0$. Then we could wonder if the resolution of the equation $Y = f(X)$ with Gröbner basis is easier. We performed tests on a toy example ($n=15, r=3, d=7$) and found that it was not different from the general case. Meaning that the fact that $\forall i \leq r \ L_i(X)L'_i(X) = 0$ does not induce any vulnerabilities.

We conducted our tests in the following way.

We generated a LL' private and public key. So we have $P'_1, \dots, P'_n = P_1 + Q_1, \dots, P_n + Q_n$ where

$$Q_i = \sum_j^r a_{i,j} L_j L'_j.$$

Because as testers we know all the $L_i L'_i$ we can select a vector $x = x_1, \dots, x_n$ such that $\forall i, L_i(x)L'_i(x) = 0$ we then note $Y = Y_1, \dots, Y_n = P'_1(x), \dots, P'_n(x) = P_1(x), \dots, P_n(x)$. Such Y is then the image of a vanilla HFE (the attacker cannot directly access to this induced HFE scheme) so we test if the degree of regularity of the system. $P'_1, \dots, P'_n$ is the same from the degree of regularity of the system $P'_1 - Y_1, \dots, P'_n - Y_n$ and if it's the same from $P'_1 - Y'_1, \dots, P'_n - Y'_n$ where Y'_i are randomly chosen.

With random ciphertext	with chosen ciphertext
17.3	17.03

Table 2. Time to solve a HFE LL' when the ciphertext is randomly chosen, compared with a ciphertext such that $Y = f(X)$ be such that $\forall i \leq r \ L_i(X)L'_i(X) = 0$

As we can see in Table 2 on average we take about 17sec to solve both. Each time we compared the same public key, first with a random ciphertext and then with a chosen one. We also found out that from a given public key no major differences were found, for example even if we have a time to solve that greatly differs from the average we still find close resolution time for a random ciphertext and a chosen one (for example 20.3 sec, 20.6 sec). From a theoretical perspective, this is not a great surprise as the algorithm that find a Gröbner basis do not involve the x_i 's in the first place. The algorithm of Buchberger involves the reduction of S-polynomials, even if it depends on the choice of Y

it does not depend on the choice of X . Then the degree of regularity of such system should not change hence the complexity of the resolution of the system via Gröbner Basis will not change.

4.7 Other attacks

In order to break HFE other attacks exists but none of them can achieve the same efficiency as MinRank or direct attacks. We can however mention differential attacks [FGS05] [DGS07]. These attacks were at the time efficient due to the choice of parameters that was at the time too optimistic. However for public keys of degree 2, due to the Gröbner basis and MinRank attacks, parameters are nowadays chosen to a point that differential attacks are irrelevant for our specific case. For a HFE polynomial of degree 3 (or higher), differential attacks are still important.

4.8 Why the L_i and $L_i L'_i$ must be kept secret

If the L_i and L'_i linear form are made public then some efficient attacks are possible. We present here two attacks.

- With Gröbner attacks: Sometimes all the products $L_i L'_i$ are 0 on a message sent. Then when the $2r$ values L_i are public, an attacker can make an exhaustive search on r forms L'_i or L_i among these $2r$ forms, say $L_1, \dots, L_r$, and then solve with a Gröbner basis algorithm the system of public equation plus $L_1 = 0, \dots, L_r = 0$. Here, the Gröbner algorithm will be efficient due to these extra equations and to the fact that all the perturbations $L_i L'_i$ vanished. *i.e.* from a ciphertext Y the attacker will find the corresponding cleartext X . Moreover, this attack still works even if wrong linear forms L_i are given in addition to the right L_i and L'_i . This attack can also easily be extended when a small number of $L_i L'_i$ are non zero.
- With MinRank: Once we know some L_i we can determine their kernel K_i and their image I_i . So we can find a projection Π_K such that $L_i \circ \Pi_K = 0$ or a projection Π_I such that $\Pi_I \circ L_i = 0$. Should we compose the projection Π_K it means we can perform an attack on the matrix $\mathbf{S}$ and find a rank of d instead of $d + r$. Similarly, we can compose the projection Π_I it means we can perform an attack on the matrix $\mathbf{T}$ and find a rank of d instead of $d + r$.

4.9 Why having numerous $L_i L'_i = 0$ cannot be used by the attacker

Obviously, an attacker could think of targeting HFE LL' when the message sent contains a lot of $L_i L'_i = 0$. In other word, let m be a message, P be the public key of a HFE LL' and $c = P(m)$ the ciphertext and suppose that for this message m all of $L_i L'_i = 0$. The intuition would be that as the message can be easily deciphered and is like a message sent from a “vanilla” HFE.

However, as we saw in previous sections MinRank attacks only attack the public key and do not take into account messages sent. We can then discard this

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attack immediately. On the other hand, direct attacks could be more threatening as their goal is to invert the system for one message and not recovering the whole key. But as we showed in previous sections, Gröbner basis attacks are not more effective against a ciphertext with a lot of $L_i L'_i$.

Furthermore, we conducted tests to evaluate the degree of regularity of the polynomials of the public key $P' = P'_1, \dots, P'_n$ and compared them with the degree of regularity of $P'_1 - y_1, \dots, P'_n - y_n$ where $y = y_1, \dots, y_n$ is chosen such that all $L_i(x_1, \dots, x_n) L'_i(x_1, \dots, x_n) = 0$. In this case, $y_i = P'_i(x_1, \dots, x_n) = P_i(x_1, \dots, x_n)$.

Parameters	Dreg	Dreg chosen ciphertext	Dreg random ciphertext
$n, q, D, LL = 12, 3, 11, 3$	13	13	13
$n, q, D, LL = 10, 3, 11, 3$	10	10	10

Table 3. Degree of regularity when choosing the ciphertext and when having a random one

It doesn't show any fall in degree of regularity. It tends to show that the complexity to find the Gröbner basis of the system does not change with the ciphertext even in the supposed worst case.

5 Number of encryption in function of the number of copies sent

In this section we will answer the question of the average number of messages (or number of session keys) required to be able to decrypt at least one of messages sent depending of the number of $L_i L'_i = 1$ allowed.

In the following we will keep the parameters we obtained in section 4.4 to avoid MinRank attacks *i.e.*:

- $r = 12$ for 80 bits of security
- $r = 17$ for 100 bits of security
- $r = 24$ for 128 bits of security
- $r = 40$ for 192 bits of security
- $r = 56$ for 256 bits of security

5.1 With all $L_i L'_i = 0$

The probability to have all $L_i L'_i = 0$ on a given encryption is $(3/4)^r$. Therefore, we will send about $(4/3)^r$ messages if we want this to occur with a good probability. For example, for 128 bits of expected security we have $r=24$ this gives approximately 996 messages to be send. In this case the number of decryptions will be equal to the number of message send, so 996 decryption (cf. Table 9)

5.2 With only one $L_i L'_i = 1$

The probability to have exactly one $L_i L'_i = 1$ is $(3/4)^{r-1} * (r/4)$. Therefore it is about $(3/4)^r + (3/4)^{r-1} * (r/4)$ for at most one $L_i L'_i = 1$. For 128 bits of security ($r=24$) this gives a probability of about 0.9% and we will send about 110 messages. In this case the number of decryptions will be about $110 + 110 \times 24 = 2750$ (cf. Table 9).

5.3 General formula

The probability to have exactly α equalities $L_i L'_i = 1$ is given by the binomial distribution:

$$\left(\frac{3}{4}\right)^{r-\alpha} \left(\frac{1}{4}\right)^{\alpha} \binom{r}{\alpha}.$$

This leads to the Tables 7,8,9,10,11 in appendix.

6 Examples of parameters and performances with public equations of degree 2

In this section we will present some explicit examples for our scheme HFE LL' with parameters and an evaluation of the performance.

We will present parameters for five level of expected security: 80, 100, 128, 192, 256 bits of security. We realised our tests of performances using the code for GeMSS (more precisely the reference implementation as it allows for flexible parameters choices). Although the code was designed for signatures, it gives us a reasonable indication for the time to decrypt (as the process of decryption is similar to the one to sign). The code does not directly model the perturbation LL' (and we cannot modify it for our own purposes) so we multiply the number of cycles we obtained by the average number of decryption required for each level of security. This gives us an evaluation of the performance that gives us a good idea of the order of magnitude of the number of cycles required to decrypt our system.

We chose the parameters in order to be protected from the attacks we previously presented.

For clarity we have chosen here only one parameter for the maximal number of $L_i L'_i = 1$ but as shown in the tables of section 5 other trade-off are possible. Here, we have chosen a HFE central map with public key of degree 2 and a secret polynomial of degree 17. With such central map the rank, without perturbation, is 5 (and not 0), it explains why in Tables 4 and 5 we used a parameter r slightly smaller than in section 5.

So we have to obtain parameters such that:

$$2^\lambda \leq \left((d+r)(n-1)^4 \binom{2(d+r)+1}{d+r}^2 \right).$$

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Where λ is the security parameter and such that:

$$2^\lambda \leq \binom{n + d_{reg}}{n}^\omega,$$

with

$$d_{reg} \approx \frac{d + r}{2} + 2.$$

Security bits	(q, n, D, r, e)	Decrypt one message (Cycles)	Avg. nbr. of decryptions	Total cycles	Total time (s)
80	(2, 100, 17, 10, 0)	2M	17.7	35.5M	0.011
100	(2, 110, 17, 14, 0)	2M	56	112M	0.035
128	(2, 138, 17, 22, 1)	2M	1700	3400M	1.06
192	(2, 200, 17, 38, 1)	3M	168883	506649	158.32
256	(2, 266, 17, 52, 2)	4M	28530507	114122030	35663.13

Table 4. Computation time of a HFE LL' scheme, q is the characteristic, n the size of the field extension, D the degree of the central map, r the number of $L_i L'_i$, and e the maximal number of $L_i L'_i = 1$. The total time is an estimation.

Security bits	(q, n, D, r, e)	$ pk $ (KB)	$ ciphertext $ of one ciphertext(b)	number of messages	Total size
80	(2, 100, 17, 10, 0)	45	100	17.7	1770.1
100	(2, 110, 17, 14, 0)	61	110	56.12	6173.2
128	(2, 138, 17, 22, 1)	163	138	73	10198.2
192	(2, 200, 17, 38, 1)	497	200	4330	866066.0
256	(2, 266, 17, 52, 2)	1171	266	20689	5503274

Table 5. Sizes of the ciphertext send for a HFE LL' schemes, q is the characteristic, n the size of the field extension, D the degree of the central map, r the number of $L_i L'_i$, and e the maximal number of $L_i L'_i = 1$.

In Table 4 we can see that up until 128 bits we have reasonable decryption time. Benchmarking was done on an Intel Core i7-10850H 3.2GHz CPU with 32GB of RAM. We can note that the computations are easily parallelisable. We could also use a different central map than HFE. For example we could use a C^* ([MI88]) central map, that is known to be faster than HFE. However, as we already mentioned, all attempts to repair C^* have failed. Therefore we are much more confident with HFE LL' than with C^* LL' where a specific analysis would be required.

As far as we know the cryptanalysis of HFE LL' does not differ much from the results obtained for HFE $\hat{+}$ as we showed earlier. Hence, we are confident with the current parameters according to the current state of attacks as we can

level of security	MinRank	Direct attacks	Hybrid
80	84	80.9	83
100	101	103	105
128	134	129	137
192	200	199	209
256	258	252	272

Table 6. Complexity of MinRank attack and direct attacks on our parameters (the complexity of Gröbner attack is optimistic)

see in Table 6 (the complexity of the attack is computed using the parameters as stated in Table 4). Unless a new idea is discovered and a vulnerability exploits the particular form of the polynomial we add, we should not expect any major changes for our parameters.

7 Conclusion

Very few efficient and secure multivariate encryption algorithms exist at present. This is mainly due to the fast progress of MinRank attacks (to recover the secret key) or Gröbner attacks (to inverse the system). New type of attack (MinRank on S , [TPD21]) and new method of resolution (support minor [BBB⁺22]) have threatened the security of HFE and also threatened the security of most multivariate encryption algorithm (many of them are based on variants of HFE). For example, Vanilla HFE is not efficient when we want more than 80 bits of security.

In this paper, we have presented a new perturbation, called LL' , to enforce the security of a multivariate encryption scheme, for example HFE. LL' perturbation was previously used only in signature schemes, and in a very different way [GP06]. With our perturbation LL' we have been able to design efficient schemes up to 128 bits of security. (For 128 bits of expected security we need about 1s to decrypt with public equations of degree 2, and the encryption is very fast, as usually in multivariate cryptography). It may also be possible to have more security bits (by using public equations of degree 3 or 4), but then the public key become very large. It could also be possible to use C^*LL' instead of $HFE_{LL'}$ since the Matsumot-Imai C^* is very fast this variant looks promising. However, so far all the attempt to repair C^* have failed. Therefore, we have to be cautious about C^*LL' and further analysis is required.

Due to the recent introduction of LL' algorithm, we do not recommend using these schemes right now since many multivariate scheme have been broken in the past. However it is interesting to notice that new ideas can still be found to reinforce the security of multivariate scheme. It is also interesting to notice that, again, it seems possible to use multivariate schemes for encryption.

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7. CONCLUSION

Computations and Transmissions for 80 bits of expected security			
Number of $L_i L'_i = 1$	Probability for 1 message sent	number of messages sent	number of decryption
0	3.16%	31.5	31.5
≤ 1	15.8 %	6.3	81.9
≤ 2	39.0 %	2.5	198
≤ 3	64.8 %	1.5	450
≤ 4	84.1 %	1.2	820
≤ 5	94.4 %	1.06	1680

Table 7. Maximal numbers of $L_i L'_i = 1$, number of required messages, number of decryption needed, security 80 bits: $r = 12$

.1 Tables of complexities for decryption with quadratic public equations

Table 7 shows orders of magnitude for the number of messages required to be sent and the number of decryption required in function of the number of accepted $L_i L'_i = 1$.

For example with $\hat{+}$ we would have required 2^{12} tries but on average we require only half of them to find the right perturbation.

In our example if we send 31 cypher text we have 63% of chance that one of the messages have all of its $L_i L'_i$ to be null. And if this happens we will find it in approximately 15-16 tries.

For 80 bits of security, the advantage of LL' against $\hat{+}$ is that we decrypt 130 times faster while retaining the same security. However, one must take into account the cost of the encryption/decryption using a secret key algorithm and the fact that one must send more cypher text in case of a failure of the decryption. The advantage is relatively limited but it will be much more significant when we increase the number of security bits.

Computations and Transmissions for 100 bits of expected security			
Number of $L_i L'_i = 1$	Probability for 1 message sent	number of messages sent	number of decryption
0	0.75%	133	133
≤ 1	5.76 %	17.3	311
≤ 2	17.12 %	5.8	794
≤ 3	36 %	2.77	1886
≤ 4	58.1 %	1.72	4093
≤ 5	77.3 %	1.29	8007

Table 8. Maximal numbers of $L_i L'_i = 1$, number of required messages, number of decryption needed, security 100 bits: $r = 17$

In Table 8 the number of decryption for $\hat{+}$ would be $2^{17} = 131072$. So if we send only 17 cypher text we will be 420 times ($131072/311 = 420$) faster than $\hat{+}$.

7. CONCLUSION

Computations and Transmissions for 128 bits of expected security			
Number of $L_i L'_i = 1$	Probability for 1 message sent	number of messages sent	number of decryption
0	0.1%	996	996
≤ 1	0.9 %	110	2750
≤ 2	3.9 %	25	7525
≤ 3	11.5 %	8.7	20227
≤ 4	24.6 %	4	51804
≤ 5	42.15 %	2.3	128000
≤ 6	60.6 %	1.6	304000

Table 9. Maximal numbers of $L_i L'_i = 1$, number of required messages, number of decryption needed, security 128 bits: $r = 24$

In Table 9, if we send about 25 cypher text the computing time will be 2200 ($2^{27}/7525$) times faster than $\hat{+}$. The advantages are then significant in regard to $\hat{+}$. One must do 7500 decryption which remains realistic. Then 128 bits of security seems reasonable according to the current complexity of the MinRank and Gröbner attacks.

Computations and Transmissions for 192 bits of expected security			
Number of $L_i L'_i = 1$	Probability for 1 message sent	number of messages sent	number of decryption
0	0.001%	100000	100000
≤ 1	0.014 %	7000	285 000
≤ 2	0.101 %	984	800 000
≤ 3	0.47 %	213	$2^{21.2}$
≤ 4	1.6 %	62	$2^{22.7}$
≤ 5	4.3 %	23	$2^{24.1}$
≤ 6	9.6 %	10.4	$2^{25.5}$
≤ 7	18.1 %	5.5	$2^{26.9}$
≤ 8	29.9 %	3.3	$2^{28.4}$

Table 10. Maximal numbers of $L_i L'_i = 1$, number of required messages, number of decryption needed, security 192 bits: $r = 40$

In Table 10 the computation will be about 4 millions times faster than $\hat{+}$. Still it requires about 285 000 decryptions. This begins to be quite excessive to be realistically used.

In Table 11, the gains compared to $\hat{+}$ are astonishing (8 billions times faster in the case of all $L_i L'_i = 0$) but it is not realistically feasible to send 10 millions messages, or to do 8 millions decryptions. And if we send 462 messages, while the gain is still 33 millions times faster it requires to decrypt 2 billions times, which is too much. Then the 256 bits of security will probably not be attained with the LL' method.

7. CONCLUSION

Computations and Transmissions for 256 bits of expected security			
Number of $L_i L'_i = 1$	Probability for 1 message sent	number of messages sent	number of decryption
0	0.00001%	10^7	2^{23}
≤ 1	0.0002 %	$5 * 10^5$	$2^{24.8}$
≤ 2	0.0019 %	52000	$2^{26.4}$
≤ 3	0.012 %	8200	2^{28}
≤ 4	0.0058 %	1700	$2^{29.4}$
≤ 5	0.216 %	462	$2^{31.0}$
≤ 6	0.665 %	150	$2^{32.4}$
≤ 7	1.73 %	57	$2^{34.0}$
≤ 8	3.91 %	25.5	$2^{35.1}$

Table 11. Maximal numbers of $L_i L'_i = 1$, number of required messages, number of decryption needed, security 256 bits: $r = 56$

.2 Variant with equation of degree 3

If the public equation are of degree 3 (instead of 2) then we can add some perturbations $L_i L'_i L''_i$ (instead of $L_i L'_i$) where L_i , L'_i and L''_i are secret linear forms in $x_1, \dots, x_n$. Then the public key will be bigger. For example with $n \approx 200$ the size of the public key in degree 3 will be about $\frac{n^4}{6}$ bits, *i.e.* about 32 Megabytes (Compared to $\frac{n^3}{2}$, about 500 kilobytes in degree 2). However, $L_i L'_i L''_i = 0$ with a probability $\frac{7}{8}$ (instead of $\frac{3}{4}$ for $L_i L'_i = 0$ in degree 2). Therefore we can expect a more efficient scheme in terms of computations and number of bits to send.

.3 Differential attack and number of terms $L_i L'_i L''_i$

Very often, when we study a multivariate scheme of degree 3, after a differential we obtain a similar scheme of degree 2. Hence, very often the study of multivariate degree 3 scheme does not fundamentally change from the one in degree 2.

Let's see the results we obtain after a differential. Let $X = (x_1, \dots, x_n)$ be and $A = (a_1, \dots, a_n)$. Let $f(x) = x_1 x_2 x_3$ be (it's a simplified notation of $L_i L'_i L''_i$). Then the differential $f(X + A) + f(A)$ is:

$$\begin{aligned} & (x_1 + a_1)(x_2 + a_2)(x_3 + a_3) + x_1 x_2 x_3 \\ &= a_1 x_2 x_3 + a_2 x_1 x_3 + a_3 x_1 x_2 \end{aligned}$$

(we ignore terms of degree 1). If $a_1 = a_2 = a_3 = 0$ then the rank of this expression is 0. If only one of the a_i is non zero, then the rank is 1. If two are non zero, then the rank is also 1, since $x_2 x_3 + x_1 x_3 = x_3(x_2 + x_1)$. If $a_1 = a_2 = a_3 = 1$ then the rank is 2, since $x_2 x_3 + x_1 x_3 + x_2 x_1 = x_2 x_1 + x_3(x_2 + x_1)$. Therefore, on average, the rank will be:

$$0 \times \frac{1}{8} + 1 \times \frac{6}{8} + 2 \times \frac{1}{8} = 1.$$

7. CONCLUSION

This means that if we use r terms $L_i L'_i L''_i$, then in the differential on average we will obtain r terms $M_i M'_i$ with M_i and M'_i of degree one. Therefore, we should keep in degree 3 about the same value r as before in degree 2 for the same level of security.

For example for 128 bits of security we will keep $r \approx 24$, as we can see in the Table 12. This Table 12 can be compared with Table 9 (in degree 2).

Computations and Transmissions for 128 bits of expected security in degree 3			
Number of $L_i L'_i L''_i = 1$	Probability for 1 message sent	number of messages sent	number of decryption
0	4.05%	24.6	24.6
≤ 1	17.9 %	5.56	139
≤ 2	40.1 %	2.45	737
≤ 3	64 %	1.56	3627

Table 12. Maximal numbers of $L_i L'_i = 1$, number of required messages, number of decryption needed, security 128 bits: $r = 24$ (degree 3)

For example in Table 12 we can send 25 messages and only 25 decryptions will be needed (instead of 7525 decryptions on 25 messages in Table 9). The only drawback would be the size of the public key of about 32 Megabyte, if $n \approx 200$ instead of about 500KB in degree 2. Moreover, the study of systems of degree 3 is rather poor compared to the prolific study of degree 2 system. Therefore, there may be better cryptanalysis possible.

Computations and Transmissions for 256 bits of expected security in degree 3			
Number of $L_i L'_i = 1$	Probability for 1 message sent	number of messages sent	number of decryption
0	0.056%	1768	1768
≤ 1	0.509 %	196	11172
≤ 2	0.0019 %	44	70268

Table 13. Maximal numbers of $L_i L'_i = 1$, number of required messages, number of decryption needed, security 256 bits: $r = 56$ (degree 3)

From Table 13 we see that for 256 bits of expected security, even with public equation of degree 3, the complexities are really high (probably not realistic, except in for very special usage): Public key of about 32 Megabyte ($n \approx 200$), about 1768 messages to send and about 1768 decryptions.

With public equation of degree 4 much less computations are required but the public key becomes unrealistic ($\frac{n^5}{24} \approx 1.55$ GB for $n \approx 200$). And then again, better cryptanalysis may exist in degree 4.